```
import numpy as np

image = np.array([[1, 2], [3, 4]])

x = image.flatten()

weights = np.array([[0.2, 0.3],
                    [0.4, 0.5],
                    [0.6, 0.7],
                    [0.8, 0.9]])

bias = np.array([0.1, 0.2])

output = np.dot(x, weights) + bias

print("Flattened Image:", x)
print("Layer Output:", output)
```

```
Flattened Image: [1 2 3 4]
Layer Output: [6.1 7.2]
```

```
import numpy as np

X = np.array([[90, 85, 80],
              [70, 75, 65],
              [95, 90, 88]])

W = np.array([[0.3], [0.4], [0.3]])

b = np.array([5])

output = np.dot(X, W) + b

print("Predicted Performance:")
print(output)
```

```
Predicted Performance:
[[90. ]
 [75.5]
 [95.9]]
```

```
import numpy as np

products = np.array([[1, 2, 3],
                     [2, 1, 2],
                     [3, 2, 1],
                     [1, 1, 1]])

customer = np.array([1, 2, 2])

similarity = np.dot(products, customer)

print("Similarity Scores:", similarity)

recommend = np.argsort(similarity)[::-1]

print("Recommended Products:", recommend + 1)
```

```
Similarity Scores: [11  8  9  5]
Recommended Products: [1 3 2 4]
```

```
import numpy as np

X = np.array([[1, 2],
              [3, 4]])

W = np.array([[0.2, 0.3],
              [0.4, 0.5]])

b = np.array([0.1, 0.2])

element = X * W

z = np.dot(X, W) + b
```

```
output = 1 / (1 + np.exp(-z))

print("Element-wise Multiplication:")
print(element)

print("Final Output:")
print(output)
```

```
Element-wise Multiplication:
[[0.2 0.6]
 [1.2 2. ]]
Final Output:
[[0.75026011 0.81757448]
 [0.90887704 0.95689275]]
```